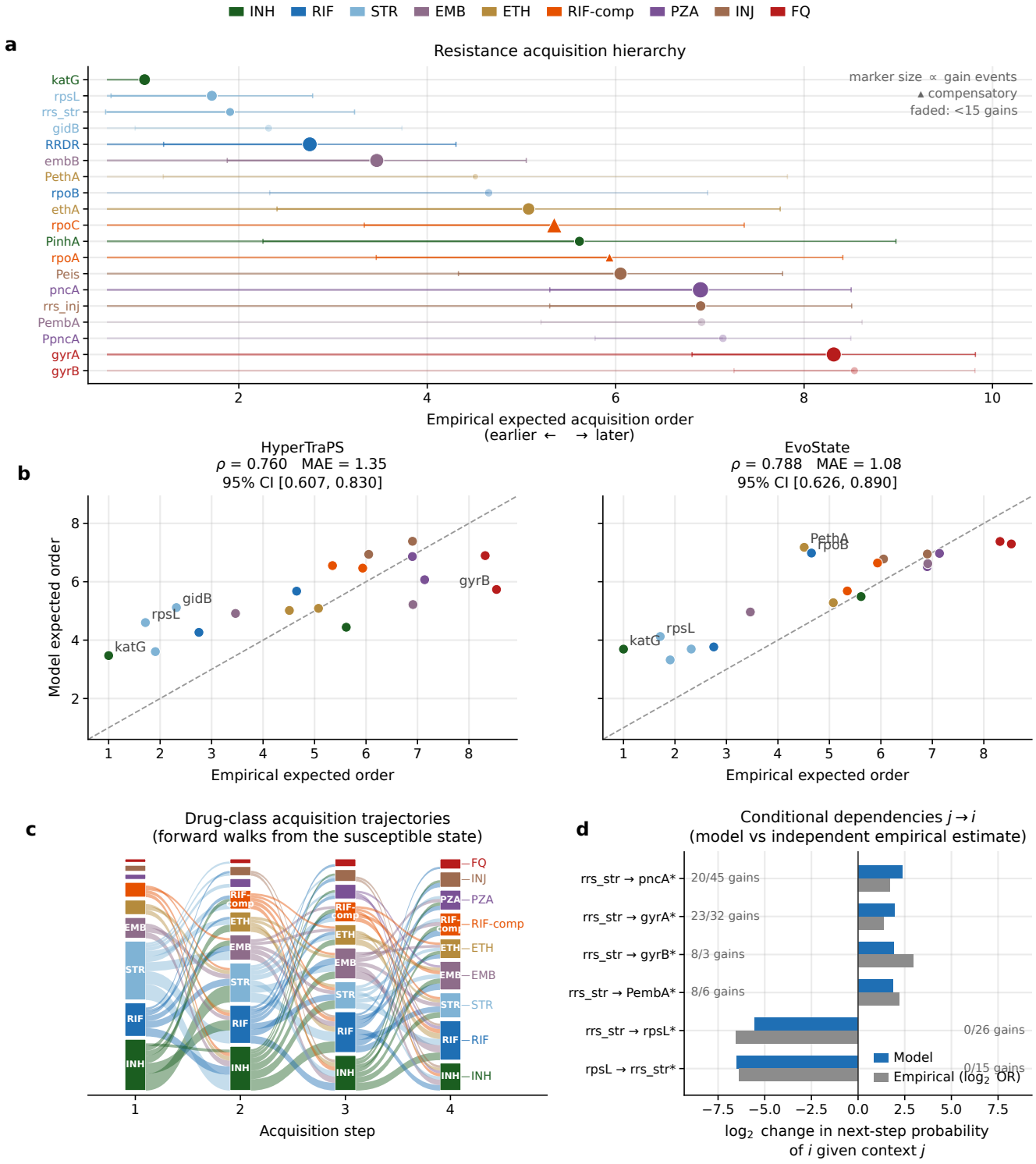
INH
RIF
STR
EMB
ETH
RIF-comp
PZA
INJ
FQ
a
Resistance acquisition hierarchy
marker size ∝ gain events
▲ compensatory
faded: <15 gains
katG
rpsL
rrs_str
gidB
RRDR
embB
PethA
rpoB
ethA
rpoC
PinhA
rpoA
Peis
pncA
rrs_inj
PembA
PpncA
gyrA
gyrB
2
4
6
8
10
Empirical expected acquisition order
(earlier ← → later)
b
HyperTraPS
ρ = 0.760 MAE = 1.35
95% CI [0.607, 0.830]
EvoState
ρ = 0.788 MAE = 1.08
95% CI [0.626, 0.890]
Model expected order
Empirical expected order
gyrB
gidB
rpsL
katG
PethA
rpoB
c
Drug-class acquisition trajectories
(forward walks from the susceptible state)
Acquisition step
d
Conditional dependencies j → i
(model vs independent empirical estimate)
rrs_str → pncA* 20/45 gains
rrs_str → gyrA* 23/32 gains
rrs_str → gyrB* 8/3 gains
rrs_str → PembA* 8/6 gains
rrs_str → rpsL* 0/26 gains
rpsL → rrs_str* 0/15 gains
Model
Empirical (log2 OR)
log2 change in next-step probability
of i given context j